Ayasha Hossain Jui

Software Engineer

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Summary

Software Engineer with 3+ years of experience designing and delivering scalable, production-grade backend systems in the FinTech domain. Proficient in Java, Spring Boot, Spring Security, and Spring Data JPA (Hibernate), with hands-on experience building concurrency-safe transaction and payment workflows (pessimistic row locking, idempotency keys, atomic balance updates, real-time UI updates via SSE) and event-driven systems for real-time market data using Kafka and RabbitMQ. Experienced with PostgreSQL, SQL Server, MySQL, and MongoDB, as well as Docker, CI/CD, and JWT/OAuth2 authentication. Ownership-driven, with a strong focus on clean code, sound architectural decisions, and cross-team knowledge sharing.

Experience

Software Engineer March 2023 – Present

Dohatec New Media – Independent Software Vendor *Dhaka, Bangladesh*

- Develop and maintain production microservices in **Java Spring Boot** for a live securities trading platform; set up the service registry, config server, and API gateway for a scalable, maintainable architecture.
- Built the end-to-end **financial service** for PQCal Global, covering **deposits (including cheque-slip deposits), withdrawals, and buying and selling of instruments**, with **atomic, idempotent, and concurrency-safe** balance updates using database transactions, **pessimistic row locking** , and **idempotency keys** to prevent duplicate or conflicting concurrent requests; pushed payment and balance updates to the UI in real time using **Server-Sent Events (SSE)**.
- Took over the **market data service** for PQCal Global mid-development; consume CME market data from **Kafka** topics and expose **RESTful APIs** that serve live views in the Angular UI.
- Developed versioned **RESTful APIs** for user management, authentication, financial services, and market analysis modules; followed clean code standards and took part in code reviews.
- Wrote and optimized queries for **SQL Server** and **PostgreSQL** using **Spring Data JPA / Hibernate**; troubleshot and tuned slow queries and APIs to support high-volume financial data processing.
- Implemented asynchronous, event-driven inter-service communication with **RabbitMQ** and **Kafka**, improving decoupling and reliability.
- Implemented authentication and authorization with **Spring Security** and **JWT/OAuth2**; extended it into a dedicated auth server for international product expansion.
- Developed reusable **Angular** UI components with PrimeNG and secure token-based authentication flows for full-stack delivery.
- Worked with QA and DevOps on **CI/CD pipelines** and **Docker** deployments; contributed to unit and functional test coverage, system design reviews, and Agile sprint planning.

Projects

PQCal (*Live Production*) Dohatec New Media

Spring Boot, FastAPI, Angular, PrimeNG, SQL Server, PostgreSQL, RabbitMQ, Docker *pqcal.dohasecurities.com.bd*

Dohatec's first in-house product — a live securities trading platform used in the Bangladeshi market.

- Contributed to the **user-management service** (BO request search, unfinished-registration filtering and cleanup, trader-client listing with pagination, commission-update propagation over RabbitMQ), the **market-analysis service** (Python, FastAPI: company fundamentals, dividend, share-holding, quarterly EPS and NAV history, index and minute-chart APIs), and the **Angular/PrimeNG web UI** (BO registration wizard with nominees and documents, position and portfolio reports, transfers, bonus shares).
- Developed paginated RESTful APIs with Spring Boot and Spring Data JPA; worked with SQL Server and PostgreSQL.
- Worked on RabbitMQ messaging and the Angular/PrimeNG UI with JWT-based authentication flows.

PQCal Global (*Under Development*) Dohatec New Media

Java Spring Boot, Angular, Kafka, RabbitMQ, SQL Server, PostgreSQL, CME

International expansion of PQCal for markets outside Bangladesh, with live paper trading via CME.

- Built the **financial service** for payment and trading flows (deposits, withdrawals, buying and selling instruments), keeping balances correct under concurrent load with **pessimistic row locking and idempotency keys** and streaming updates to the Angular UI via **SSE**.

- Took over the **market data service** mid-development: consume CME market data from **Kafka** topics and expose **REST APIs** that power live views in the Angular UI for global users.
- Consume **CME (Chicago Mercantile Exchange)** market data from Kafka to power real-time paper trading for global users.
- Set up the microservices infrastructure (service registry, config server, API gateway) with **RabbitMQ** for event-driven communication.
- Built the **user-management service**, **authorization server** (OAuth2/JWT with a custom grant), **business-rule configuration service** and **file-management service**, and contributed Angular features across the web app.

Dohatec AI (*RAG Application*) Dohatec New Media

Angular, Spring Boot RAG backend (team-built), LLM

Company AI assistant, built by a cross-functional team, that answers questions from users' own documents through a chat interface.

- Built **Angular** frontend features for document management and the chat interface, connecting users to the RAG backend; collaborated through code reviews and sprint planning.
- Worked with the backend developers on the **RAG pipeline** flow (document ingestion, chunking, embeddings, semantic retrieval, and **LLM** answer generation), contributing technical analysis to help diagnose and resolve backend issues.

Biblioteca (*Personal Project*) github.com/ayashaJui/book_tracking

Java 21, Spring Cloud, OAuth2/JWT, PostgreSQL, Angular, RxJS, Docker

Book-tracking platform (catalogue of books, authors, publishers and series; personal library and reading preferences) built as independent microservices.

- Architected independent **Spring Boot microservices** (user, catalog, user-library) with a database per service, **Eureka** service discovery, **Spring Cloud Config**, and a **Spring Cloud Gateway** as the single entry point.
- Built a dedicated **OAuth2 Authorization Server** issuing **JWT** tokens, with resource servers validating them and the Angular SPA using the authorization code flow.
- Managed schemas with **Flyway** migrations, documented APIs with **OpenAPI/Swagger**, and containerized the registry, config, gateway, auth and user services with **Docker Compose**.
- Built the **Angular 20** frontend with PrimeNG, Tailwind CSS and RxJS services across 14 lazy-loaded feature modules, with an HTTP interceptor attaching the bearer token.

StudyNest (*Live*) studynest-z0r3.onrender.com

Next.js, NestJS, TypeScript, MongoDB, JWT, Swagger, GitHub Actions

Interview-prep platform with nested topics, markdown notes, flashcards, quizzes, and a coding-problem tracker.

- Built a full-stack app with a **Next.js** frontend and a separate **NestJS** REST API on **MongoDB** (migrated from Next.js API routes), documented with Swagger and deployed as two services on Render.
- Implemented **SM-2 spaced-repetition** scheduling for flashcards and interval-based review scheduling for coding problems, plus a due-today review queue, timezone-aware study streaks, and global search.
- Secured the app with **HttpOnly cookie-based JWT** auth, bcrypt password hashing, and email password reset; added CSV/Markdown import and export.

Ink & Paper inkandpaper-frontend.onrender.com

React, Redux, Node.js, Express.js, MongoDB, Material UI

- Built a full-stack e-commerce bookstore with user authentication, book browsing, author and review exploration, and a shopping cart.
- Designed RESTful APIs with Express and MongoDB; built a responsive React UI using Material UI and Redux for global state.

TourVille (*University Project*) github.com/ayashaJui/CSEKU_WPL_2020_tourville

PHP, MySQL, JavaScript, Bootstrap, jQuery, Stripe

Tourism management platform for discovering, booking, and managing travel packages, built with a teammate for the Web Programming Lab at Khulna University.

- Built a multi-role system for **tourists, travel agencies, and administrators**: package search and browsing, agency profiles, bookings with Budget, Comfortable, and Luxury pricing tiers, reviews, and admin approval of accounts.
- Integrated **Stripe** payments, letting tourists pay a booking in full, in half, or as a percentage deposit.
- Implemented role-based agency accounts (Owner, Manager, Staff) and an agency analytics dashboard with **Google Charts**; used **PDO prepared statements** and bcrypt-hashed passwords.

Skin Cancer Classification (*Research Project*) IEEE ICCIT 2022

Python, TensorFlow, Pandas, NumPy, CNN, Xception, Transfer Learning, ISIC 2019

Deep-learning approach to classify skin cancer from images; published at the 25th International Conference on Computer and Information Technology (ICCIT).

- Fine-tuned the pre-trained **Xception** CNN with **transfer learning** to classify **eight types of skin cancer**, using **data augmentation** to add diversity during training.
- Evaluated the model on the **ISIC 2019** dataset, reporting better results than comparable studies.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, PHP, C++, SQL

Problem Solving: 350+ LeetCode problems solved in Java, Python, and C++ (arrays, strings, hash tables, binary search, dynamic programming, backtracking)

Backend: Spring Boot, Spring Cloud (Gateway, Eureka, Config), Spring MVC, Spring Security, Spring Data JPA / Hibernate, Flyway, FastAPI, Node.js, Express.js, NestJS

Frontend: Angular (NgRx, PrimeNG, Tailwind CSS), React (Redux, Material UI), Next.js, HTML, CSS, jQuery

AI & GenAI: Retrieval-Augmented Generation (RAG), LLM Integration, Document Embeddings, Semantic Search, Text Chunking

Machine Learning: TensorFlow, Pandas, NumPy, Convolutional Neural Networks (CNN), Transfer Learning (Xception), Data Augmentation, Image Classification

Databases: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB

Messaging & Streaming: Kafka, RabbitMQ, Server-Sent Events (SSE)

DevOps & Tools: Docker, Docker Compose, Git, GitHub Actions, CI/CD pipelines, Swagger/OpenAPI, IntelliJ IDEA, Tomcat

Practices: OOP, Design Patterns, Clean Code, JUnit/Mockito, REST API Design & Versioning, Microservices, Event-Driven Architecture, Real-Time Data Processing, Concurrency-Safe Transactions, Pessimistic Locking, Idempotency, JWT/OAuth2, API Gateway & Service Discovery, Debugging & Troubleshooting, Agile/Scrum, Code Reviews

Education

Khulna University — Khulna, Bangladesh
B.Sc. Engineering in Computer Science and Engineering — CGPA: 3.72/4.00 — *2018 – 2023*

Publications

Ayasha Hossain Jui, Samia Sharnami, Aminul Islam. *"A CNN Based Approach to Classify Skin Cancers using Transfer Learning"*, 25th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 17–19 December 2022. DOI: 10.1109/ICCIT57492.2022.10055838